

Solstice optimizes thermal growing season

Supplementary figures

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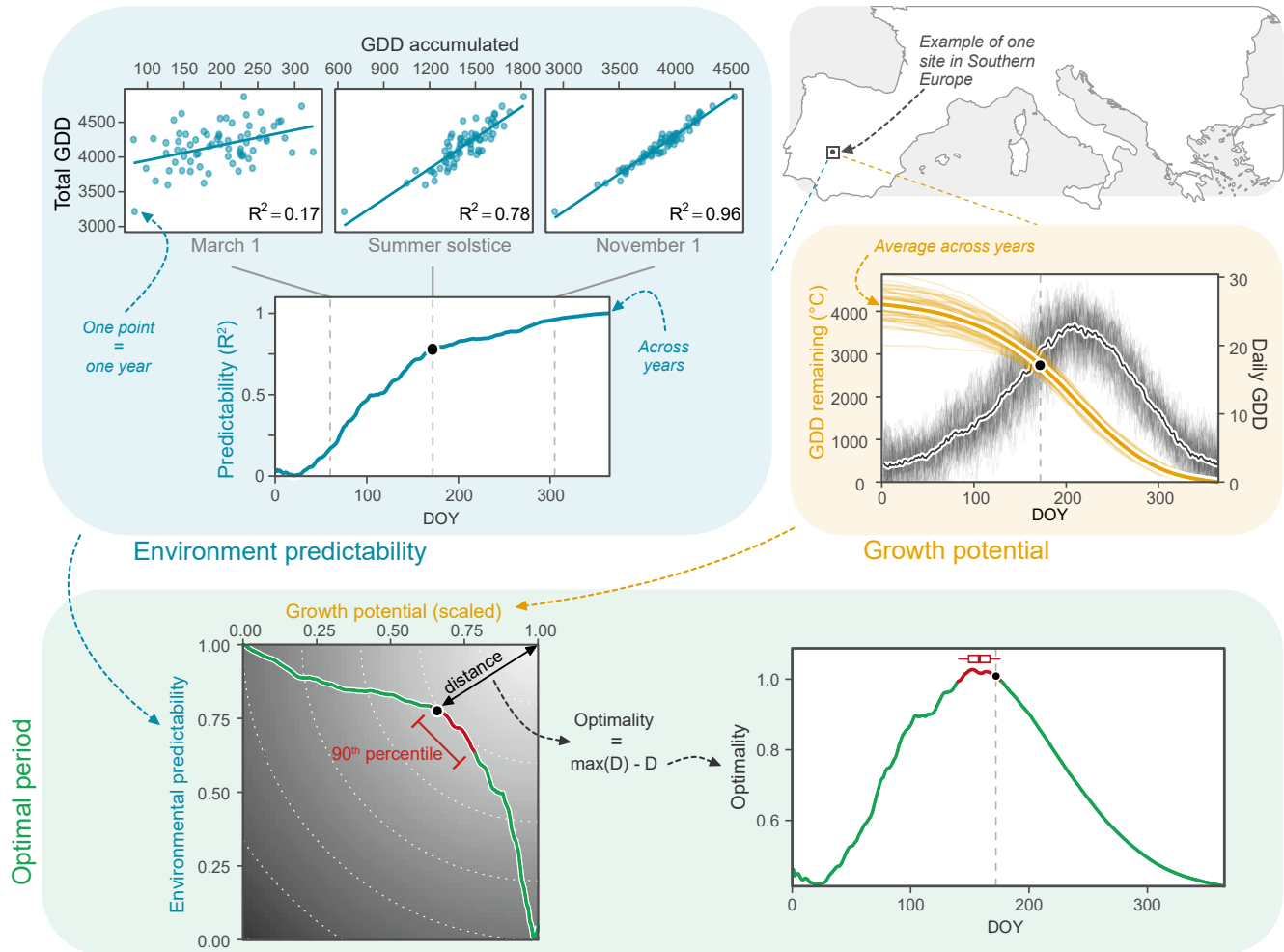


Figure S1: Workflow of the optimality analysis for one site in Southern Europe.

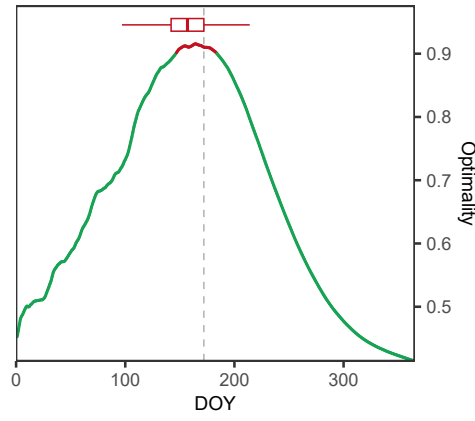


Figure S2: Average optimal trade-off between environmental predictability and growth potential across Europe, with an alternative GDD definition ($T_{lower} = 0^{\circ}\text{C}$ and $T_{upper} = 40^{\circ}\text{C}$).

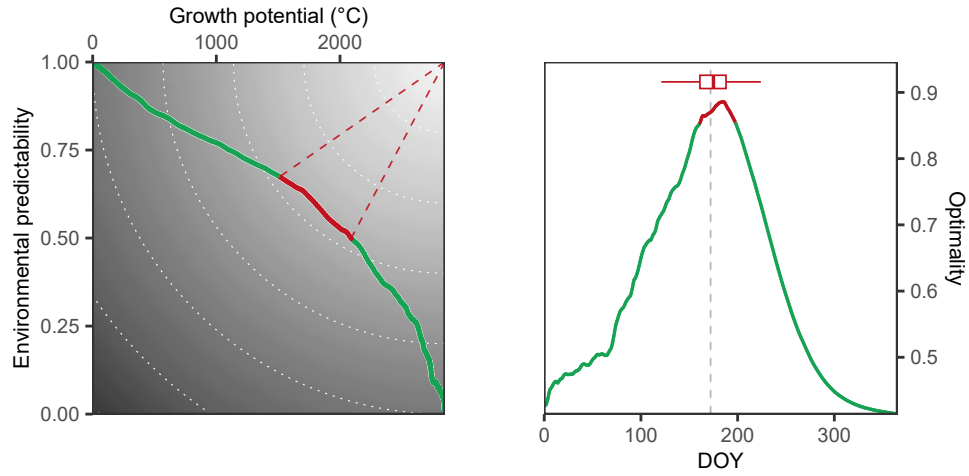


Figure S3: Solstice also marks the average optimal trade-off between environmental predictability and growth potential across North America (1951-2020). The red sections of the green curves represent days where optimality falls within the 90th percentile (i.e. top 10% most optimal days).

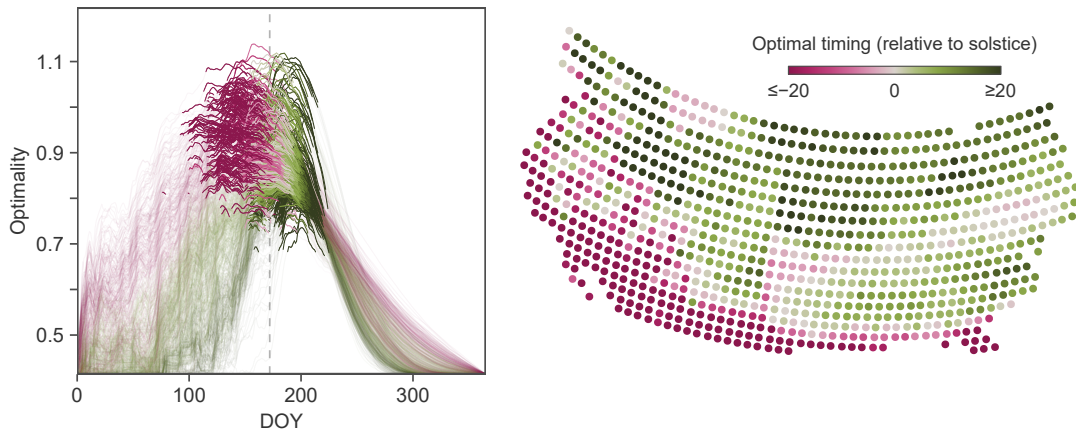


Figure S4: Average optimal timing (Figure S3) hides variation in optimal timing across the different climatic conditions of North America (1951-2020). On the left panel, each curve shows the optimality for a given site. Sites are sampled on a regular grid across Europe, as shown on the central map. Colors indicate the timing—relative to the solstice—of the median optimal day.

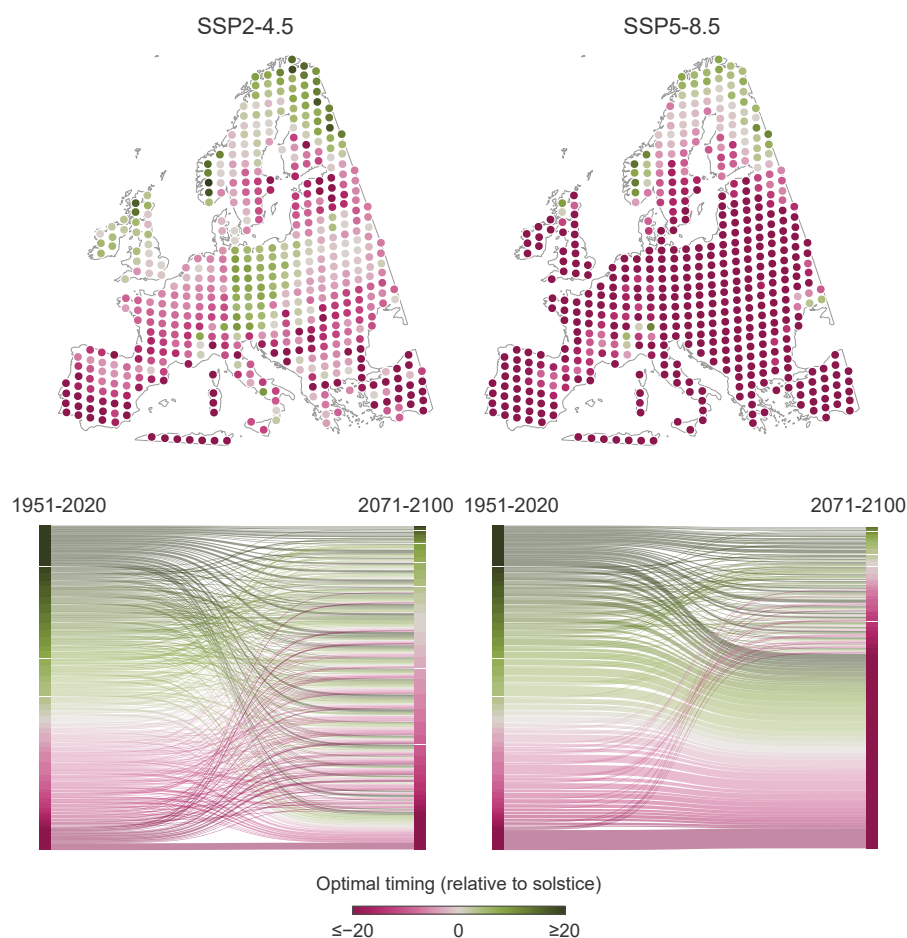


Figure S5: Average optimal timing (??) hides variation in optimal timing across the different future climatic conditions of Europe (2071-2100). On the top panel, sites are sampled on a regular grid across Europe. On the bottom panel, Sankey diagrams show the proportional flow of (site-specific) optimal timing from 1951-2020 to 2071-2100. Colors indicate the timing—relative to the solstice—of the median optimal day.